Sayde L. King, Ph.D.

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EDUCATION

Doctor of Philosophy in Computer Science and Engineering August 2025
University of South Florida, Tampa, FL
Dissertation Title: *An Exploratory Analysis of Automated Deception Detection for Mental Health Applications*
Advisor: Tempestt Neal

Master of Science in Computer Science May 2022
University of South Florida, Tampa, FL

Bachelor of Science in Computer Science May 2019
University of South Florida, Tampa, FL
Minor: International Studies

ACADEMIC POSITIONS

NASEM National Research Council Postdoctoral Research Associate Sept. 2025–Present
711th Human Performance Wing, Air Force Research Laboratory, Dayton, OH

- Lead human-centered AI research on user trust, transparency, explainability, deep learning, machine learning, and large language models.
- Design empirical studies integrating qualitative methods and AI modeling approaches to evaluate human interaction with AI-enabled systems.
- Analyze qualitative and behavioral datasets to generate actionable insights for Air Force-relevant human performance research.
- Disseminate findings through peer-reviewed publications, technical reports, and conference presentations.

PUBLICATIONS

Peer-Reviewed Journal Articles

1. S. A. Jessup, A. Capiola, G. M. Alarcon, S. M. Willis, D. Johnson, **S. L. King**, S. K. Myers, and T. Vonderhaar, "'Show Me and Tell Me!' Exploring the Effects of Modality Explanations from (In)Accurate Machine Learning Algorithms," *Behaviour & Information Technology*, 45 (5), 2026. *(Under Review).*
2. **S. L. King**, S. Bhaskar, K. Kosyluk, and T. Neal, "Ethics and Fairness Considerations in AI-Based Deception Detection Technologies for Mental Health Applications: Focus Group Study," *JMIR AI*, 5:e86633, 2026. doi: 10.2196/86633.
3. **S. L. King** and T. Neal, "Applications of AI-Enabled Deception Detection Using Video, Audio, and Physiological Data: A Systematic Review," *IEEE Access*, vol. 12, pp. 135207–135240, 2024. doi: 10.1109/ACCESS.2024.3462825.
4. **S. King**, S. Pinder, D. Fernandez-Lanvin, C. G. Garcia, J. De Andres, and M. Labrador, "Noise Signature Identification Using Mobile Phones for Indoor Localization," *Multimedia Tools and Applications*, vol. 83, pp. 64591–64613, 2024. doi: 10.1007/s11042-023-17885-3.
5. T. Neal, A. Negro, F. Montagna, M. N. Teng, S. Thomas, **S. King**, and R. Khan, "Analysis of the Evolution of COVID-19 Disease Understanding Through Temporal Knowledge Graphs," *Frontiers in Research Metrics and Analytics*, 8:1204801, 2023.
6. K. Kosyluk, J. T. Tran, **S. King**, K. Torres, and T. Neal, "Preliminary Effectiveness Study of the Cope Notes Digital Mental Health Program," *Journal of Mental Health*, 32(3), pp. 625–633, 2023.
7. **S. L. King**, J. Lebert, L. A. Karpisek, A. Phillips, T. Neal, and K. Kosyluk, "Characterizing User Experiences With an SMS Text Messaging-Based mHealth Intervention: Mixed Methods Study," *JMIR Formative Research*, 6(5):e35699, 2022.

Peer-Reviewed Conference Proceedings

1. **S. L. King**, B. Rodgers, G. M. Alarcon, "Performance, Process, and Purpose: A Qualitative Investigation into the Drivers of Human-AI Trust", in *Proceedings of the 60th Hawaii International Conference on System Sciences*, 2027. *(Submitted).*
2. **S. L. King**, H. Abootalebi, G. N. Wai, and T. Neal, "Evaluating Visual and Behavioral Signals of Deception in Real-World Contexts," in *Social Computing, Behavioral-Cultural Modeling*, SBP-BRiMS 2025, Lecture Notes in Computer Science, vol. 16127, Springer, 2026. **Best Paper Award**.

3. **S. L. King** and T. Neal, "Exploring Vision-Based Features for Detecting Deception in Well-Being: A Cross-Domain Comparison," in *Proceedings of the 2025 19th IEEE International Conference on Automatic Face and Gesture Recognition*, Clearwater, FL, 2025.
4. **S. L. King**, M. Ebraheem, P. Dang, and T. Neal, "Toward Emotion Recognition and Person Identification Using Lip Movement from Wireless Signals: A Preliminary Study," in *Proceedings of the 2024 18th IEEE International Conference on Automatic Face and Gesture Recognition*, Istanbul, Turkey, 2024.
5. **S. L. King**, N. Johnson, K. Kosyluk, and T. Neal, "Therapist Perceptions of Automated Deception Detection in Mental Health Applications," in *Artificial Intelligence in HCI*, HCII 2023, Lecture Notes in Computer Science, vol. 14050, Springer, 2023.
6. W. Lozano, **S. L. King**, and T. Neal, "Observations of Caregivers of Persons with Dementia: A Qualitative Study to Assess the Feasibility of Behavior Recognition Using AI for Supporting At-Home Care," in *Human Aspects of IT for the Aged Population*, HCII 2023, Lecture Notes in Computer Science, vol. 14050, Springer, 2023.
7. N. Loecher, **S. King**, J. Cabo, T. Neal, and K. Kosyluk, "Assessing the Efficacy of a Self-Stigma Reduction Mental Health Program with Mobile Biometrics: Work-in-Progress," in *Proceedings of the 2023 17th IEEE International Conference on Automatic Face and Gesture Recognition*, Waikoloa Beach, HI, 2023, pp. 1–6.
8. M. Ebraheem, **S. King**, and T. Neal, "Lip Movement as a WiFi-Enabled Behavioral Biometric: A Pilot Study," in *HCI International 2022 Posters*, Communications in Computer and Information Science, vol. 1583, Springer, 2022.
9. **S. King**, M. Ebraheem, K. Zanna, and T. Neal, "Learning a Privacy-Preserving Global Feature Set for Mood Classification Using Smartphone Activity and Sensor Data," in *2020 15th IEEE International Conference on Automatic Face and Gesture Recognition*, Buenos Aires, Argentina, 2020, pp. 660–664.
10. K. Zanna, **S. King**, T. Neal, and S. Canavan, "Studying the Impact of Mood on Identifying Smartphone Users," arXiv preprint arXiv:1906.11960.

FUNDING, FELLOWSHIPS, AND HONORS

Selected Funding and Fellowships

Total selected funding: $273,000+ in competitive fellowship and scholarship support

National Research Council Research Associateship, administered by the National Academies of Sciences, Engineering, and Medicine; sponsored by the Air Force Research Laboratory
Amount: $95,000 annual stipend; total: $190,000 — 2025–2027

USF Summer Dissertation Completion Fellowship, University of South Florida
Amount: $10,000 — 2025

McKnight Dissertation Fellowship, Florida Education Fund
Amount: $13,000 — 2023–2025

GEM Employer Fellowship, National GEM Consortium; sponsored by MIT Lincoln Laboratory
Amount: $20,000 — 2022–2025

Sloan PhD Scholarship, Alfred P. Sloan Foundation
Amount: $40,000 — 2019–2025

NSF Flit-Path Cohort-B Scholarship, Florida IT Graduation Attainment Pathways — 2018–2019

James and Michelle Austin Ambassador Scholarship, University of South Florida — 2018–2019

Florida Academic Scholar, Florida Bright Futures Scholarship Program — 2014–2016

Awards and Competitive Professional Development

Best Paper Award, SBP-BRiMS — 2025

Future Faculty Program Participant, Virginia Tech — 2024

Future Faculty Career Exploration Program Participant, Rochester Institute of Technology — 2024

Academic Careers Workshop Participant, CMD-IT — 2024

Apple Polishing Awardee, University of South Florida's Ambassadors — 2022

Deep Learning Bootcamp Attendee, USF Institute for AI + X — 2022

Research Bootcamp Attendee, Florida Alliance for Graduate Education in the Professoriate Scholar — 2021

Grad Cohort for Women Workshop Attendee, Computing Research Association — 2021

Institute on Teaching and Mentoring Travel Award — 2019, 2022, 2025

Member, University of South Florida, Order of the Golden Brahman — 2017–2026

RESEARCH EXPERIENCE

Ph.D. Candidate — Fall 2019–Summer 2025
Cyber Identity and Behavior Research Lab, University of South Florida, Tampa, FL
Advisor: Tempestt Neal
AI-Enabled Deception Detection for Mental Health

- Conducted multimodal data collection to support exploratory deception detection research across video, audio, gaze, and physiological modalities and across topical domains.
- Evaluated literature on AI-enabled deception detection, human-inspired deception detection, and deception in therapeutic settings.
- Conducted a study examining the prevalence and impact of deception in therapy and perceptions of AI-enabled deception detection in therapeutic contexts.

Graduate Research Assistant — Fall 2019–Summer 2025
Cyber Identity and Behavior Research Lab, University of South Florida, Tampa, FL
PI: Kristin Kosyluk; Co-I: Tempestt Neal
Up To Me: Erasing the Stigma of Mental Illness on College Campuses; NIDILRR Award No. 90IFRE0056

- Led longitudinal data collection of behavioral smartphone sensing data and accompanying self-report mental well-being surveys.
- Developed machine learning models to detect behaviors related to student success outcomes, including sleep, sense of belonging, academic performance, and mental health.
- Created user guides and explanatory videos to support participant enrollment in the behavioral sensing component of the project.

Research Assistant — Summer 2023
Cyber Identity and Behavior Research Lab, University of South Florida, Tampa, FL
PI: Tempestt Neal
Age-Aware User Authentication; National Science Foundation Award No. 2039379

- Conducted multimodal data collection sessions capturing physiological, video, audio, mouse dynamics, keystroke, and touch data to inform age-aware continuous authentication.
- Analyzed data using machine learning and deep learning approaches to inform future experimentation.

GEM Fellow Summer Research Program Intern — Summer 2022
Massachusetts Institute of Technology Lincoln Laboratory, Lexington, MA
Secret Security Clearance

- Improved feature estimation methods for musculoskeletal injury prediction using accelerometry and LSTM modeling of ground reaction force waveforms.
- Implemented asymmetry features to identify gait anomalies and informed protocol changes for laboratory modeling of musculoskeletal injury.
- Developed future data collection opportunities with Marine Corps leadership while on fielding.

National Security Internship Program Ph.D. Intern — Summer 2021–Fall 2021
Pacific Northwest National Laboratory, Department of Energy, Richland, WA

- Applied machine learning and deep learning techniques to mass spectrometry data to learn and predict patterns across spectra, instruments, and energy.
- Surveyed literature on lidar sensors, 3D point clouds, and adversarial attacks.

Graduate Assistant — Fall 2021–Spring 2023
Department of Computer Science and Engineering, University of South Florida, Tampa, FL
Co-I: Ken Christensen
Florida IT Graduation Attainment Pathways; National Science Foundation Award No. 2130298

- Recruited academically talented undergraduates in Computer Science, Information Technology, Cybersecurity, and Computer Engineering across USF, UCF, and FIU.
- Planned scholar events supporting preparation for industry careers, graduate school, research careers, and entrepreneurship.
- Mentored scholars through resume review, research and leadership opportunities, and academic success planning.

Graduate Research Assistant Fall 2019–Spring 2022
Cyber Identity and Behavior Research Lab, University of South Florida, Tampa, FL
Co-PI: Tempestt Neal

Early Detection of Disease Outbreaks using Self-Organizing Patterns - COVID-19; National Science Foundation Award No. 2028051

- Contributed to an interdisciplinary public-private team for an NSF RAPID grant.
- Created a user-friendly knowledge graph representing diseases, treatments, and comorbidities.
- Assisted in creation and dissemination of survey materials.

Ubiquitous Sensing for Mental Health Text Messaging Interventions

- Conducted interdisciplinary research with faculty in the Department of Mental Health Law and Policy.
- Interviewed users of mobile mental health intervention services and analyzed perceptions of continuous sensing data extraction for improving mental health interventions.

Student Volunteer, Research Experience for Undergraduates June 2018–August 2018
Department of Computer Science and Engineering, University of South Florida, Tampa, FL
PI: Miguel Labrador

- Participated in an NSF-funded ubiquitous computing research program and developed an audio-based indoor localization system on a three-person team.
- Collected 25-minute audio files across 19 building areas and implemented audio processing, feature extraction, and machine learning algorithms using WEKA.

TEACHING EXPERIENCE

Graduate Teaching Associate Fall 2023–Spring 2025
Department of Computer Science and Engineering, University of South Florida, Tampa, FL

- Assessed and graded assignments for 65 graduate and undergraduate students in Smart and Connected Health and Mobile Biometrics.
- Assessed and graded assignments for 70 graduate students in Security and Privacy in Machine Learning.
- Developed mobile app demonstrations with detailed video explanations connecting demonstrations to course content.
- Provided feedback, held office hours, and used learning management systems to maintain grading records.

Graduate Teaching Assistant Fall 2019–Summer 2020
Department of Computer Science and Engineering, University of South Florida, Tampa, FL

- Assisted with instructional preparation, grading, and student support for approximately 75 undergraduate and graduate students.
- Courses: Software Systems Development; Biometric Authentication on Mobile Devices; Automata Theory and Formal Languages.

Guest Lectures

Guest Lecturer, Mobile Biometrics, University of South Florida Fall 2024
Instructor of Record: Dr. Tempestt Neal
"Decision Trees and Support Vector Machines"

Guest Lecturer, Cybersecurity and Biometrics, University of South Florida Spring 2024
Instructor of Record: Dr. Tempestt Neal
"Applied Biometrics for Automated Deception Detection"

MENTORING AND ADVISING

Scholar Mentor Fall 2021–Spring 2023
Florida IT Graduation Attainment Pathways, University of South Florida, Tampa, FL

- Provided informal academic and professional mentorship to undergraduate scholars, including resume review, research opportunity guidance, leadership opportunity guidance, and academic success support.

Mentor Fall 2020–Fall 2022
Machine Learning Club, Patel High School, Tampa, FL

- Developed lessons and curriculum introducing core machine learning concepts for students in grades 9–12.
- Established assessment tools to provide feedback to participating students and support teaching partners.

Mentor May 2018–July 2018
Bulls Engineering Youth Experience, University of South Florida, Tampa, FL

- Taught robotics and engineering concepts to middle school students from low- and middle-income communities in Hillsborough County.
- Led program projects and cultivated an environment of growth, humility, and empowerment.

PRESENTATIONS AND POSTERS

1. **S. King** and T. Neal, “An Exploratory Analysis of Automated Deception Detection for Mental Health Applications,” Doctoral Consortium, 2025 19th IEEE Conference on Automatic Face and Gesture Recognition, Clearwater, FL, May 2025.
2. M. King, M. Wilks-Otto, C. Spivey, **S. L. King**, T. Neal, and K. Kosyluk, “Reducing Internalized Stigma Among College Students Through the Up to Me Program,” USF Graduate Research Symposium, Tampa, FL, March 2025.
3. **S. L. King** and T. Neal, “An Analysis of Intra- and Inter-Domain Biometric Features for Cross-Domain Deception Detection,” Second USF Artificial Intelligence + X Symposium, Tampa, FL, October 2024.
4. **S. King**, “Qualitative and Experimental Analysis of Mental Health Clinician Experiences with Client Deception,” McKnight Annual Fellows Meeting and Research & Writing Conference, Tampa, FL, February 2024.
5. **S. King** and E. Gallagher, “Overcoming Deception in Suicide Assessment,” 14th National Update on Behavioral Emergencies Conference, Las Vegas, NV, December 2023.
6. **S. King** and T. Neal, “An Exploratory Analysis of Automated Deception Detection for Mental Health Applications,” Doctoral Consortium, 17th IEEE Conference on Automatic Face and Gesture Recognition, Waikoloa Village, HI, January 2023.
7. **S. King**, P. Dang, M. Ebraheem, and T. Neal, “Toward Emotion Recognition and Person Identification Using Lip Movement From Wireless Signals: A Preliminary Study,” USF FL-AGEP Research Symposium, Tampa, FL, July 2022.
8. M. Ebraheem, **S. King**, and T. Neal, “Lip Movement as a WiFi-Enabled Behavioral Biometric: A Pilot Study,” 24th International Conference on Human-Computer Interaction, Virtual, June 2022.
9. M. Ebraheem, **S. King**, and T. Neal, “Towards a Privacy-Preserving Emergency Detection System via Channel State Information,” 1st Annual Nelms Workshop on Women in IoT, Virtual, October 2020.

INVITED TALKS AND PANELS

Panelist, “Navigating the Postdoc Labyrinth: Strategies for Success” Oct. 2025
Sloan Scholar Career Exploration and Networking Event, Institute on Teaching and Mentoring

Keynote Speaker, “Applied Affective Computing for Mental Health” May 2024
Fourth Workshop on Applied Multimodal Affect Recognition, 18th IEEE International Conference on Automatic Face and Gesture Recognition

Presenter, “Applied Biometrics for Deception Detection” Apr. 2024
Institute for AI + X Seminar, University of South Florida

Presenter, “Introduction to Face Recognition” July 2023; May 2021
CodeBreakHERs

Panelist, 6th Annual Florida IT Graduation Attainment Pathways Symposium, University of Central Florida Apr. 2022

Guest Speaker, “Big Brain Energy” Apr. 2022
Modern Figures Podcast

Panelist, Flit-Path: Graduate School Showcase, University of Central Florida Nov. 2021

Panelist, “Women in STEM” Oct. 2020
National Society of Black Engineers

Speaker, “Is Undergraduate Research Worth It?” Sept. 2020
Women in Computer Science and Engineering

Panelist, Tampa Bay STEM Transfer: Bridge to Baccalaureate Alliance, University of South Florida Dec. 2019

Panelist, “Obstacles and Opportunities for Internships” Apr. 2019
3rd Annual Florida IT Graduation Attainment Pathways Symposium

SERVICE AND PROFESSIONAL LEADERSHIP

Peer Reviewer

Full Reviewer

International Conference on Social Computing, Behavioral-Cultural Modeling & Prediction and Behavior Representation in Modeling and Simulation — 2026
International Conference on Automatic Face and Gesture Recognition — 2025–2026
International Joint Conference on Biometrics — 2021–2026
IEEE Transactions on Affective Computing — 2025–2026
Springer Nature: Scientific Reports — 2024
Journal of Medical Internet Research: Research Protocols — 2024

Auxiliary Reviewer

International Joint Conference on Biometrics — 2020–2022
IEEE Transactions on Biometrics, Behavior, and Identity Science — 2019–2020
Workshop on Demographic Variation in the Performance of Biometric Systems, IEEE Winter Conference on Applications of Computer Vision — 2020
Challenges and Opportunities for Privacy and Security, IEEE Conference on Computer Vision and Pattern Recognition — 2019
IEEE International Conference on Biometrics: Theory, Applications, and Systems — 2019
ACM Computing Surveys — 2019

Conference and Workshop Leadership

Technical Program Committee Member — 2025
IEEE International Joint Conference on Biometrics – Osaka, Japan

Publicity Chair — 2025
19th IEEE International Conference on Automatic Face and Gesture Recognition – Clearwater, FL

Program Committee Member — 2023
First Workshop on Interdisciplinary Applications of Biometrics and Identity Science, IEEE FG – Waikoloa Beach, HI

Program Committee Member — 2020–2022
Workshop on Applied Multimodal Affect Recognition, IEEE International Conference on Pattern Recognition – Montreal, Quebec

Program Committee Member — 2020
Special Session on Identity for Social Good, IEEE International Joint Conference on Biometrics – Houston, TX

RELATED PROFESSIONAL EXPERIENCE

Systems Analyst — May 2017–August 2017
RPI Consultants, Brandon, FL

- Completed and presented deliverables including custom program and workflow development and report development.
- Researched and troubleshot in support of Senior Technical Consultants and produced technical documentation, user guides, and test scripts.

Cybersecurity Intern — January 2017–May 2017
Abacode Cybersecurity Experts, Tampa, FL

- Created white papers explaining technical cybersecurity topics for lay audiences and drafted press releases for events, cybersecurity news, and company milestones.

Technical Services College Intern — June 2015–August 2015
Northrop Grumman Corporation, Defense Manpower Data Center, Seaside, CA

- Collaborated on a two-person team to modify an administrative tool using relational databases and managed technical write-ups and weekly client presentations.